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Ionosphere dynamics over the Southern Hemisphere during the 31 March 2001 severe magnetic storm using multi-instrument measurement data

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The effects of the 31 March 2001 severe magnetic storm on the Southern Hemisphere ionosphere have been studied using ground-based and satellite measurements. The prime goal of this comprehensive study is to track the ionospheric response from high-to-low latitude to obtain a clear understanding of storm-time ionospheric change. The study uses a combination of ionospheric Total Electron Content (TEC) obtained from GPS signal group delay and phase advance measurements, ionosonde data, and data from satellite in-situ measurements, such as the Defense Meteorological Satellite Program (DMSP), TOPographic EXplorer (TOPEX), and solar wind data from the Advanced Composition Explorer (ACE). A chain of Global Positioning System (GPS) stations near the 150° E meridian has been used to give comprehensive latitude coverage extending from the cusp to the equatorial region. A tomographic inversion algorithm has been applied to the GPS TEC measurements to obtain maps of the latitudinal structure of the ionosphere during this severe magnetic storm period, enabling both the spatial and temporal response of the ionosphere to be studied. Analysis of data from several of the instruments indicates that a strong density enhancement occurred at mid-latitudes at 11:00 UT on 31 March 2001 and was followed by equatorward propagating large-scale Travelling Ionospheric Disturbances (TIDs). The tomographic reconstruction revealed important features in ionospheric structure, such as quasi-wave formations extending finger-like to higher altitudes. The most pronounced ionospheric effects of the storm occurred at high- and mid-latitudes, where strong positive disturbances occurred during the storm main phase, followed by a long lasting negative storm effect during the recovery phase. Relatively minor storm effects occurred in the equatorial region